

Title: **Timestamp-Aware Metadata- Augmented Rag**

Subtitle:
An AI Tutor for Educational Video Learning

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Introduction

- AI Tutors help students learn better.
- RAG improves answer accuracy.
- But still has some problems.

Problem Statement:

Problems In Traditional RAG:

- Fragmented context (broken information).
- Weak retrieval (wrong chunks).
- Hallucinations(AI guesses).
- No link to video source.

Inspiration:

- KG – RAG uses knowledge Graph.
- Links concepts for deeper understanding.
- Provides context-aware answers.
- But it is complex and heavy.



Our Idea:

Simple Alternative to KG-RAG

Instead of complex graph, we use:

- **Chunk Aggregation** – Merge 5 chunks for better context.
- **Metadata** - Add Video title, section info to each chunk.
- **Timestamps** – Link every answer to exact video time.

System Flow:

Video ---> Audio ---> Text ---> Chunks
---> Embedding ---> Retrieval ---> LLM
---> Answer

Step 1: Preprocessing

- **FFmpeg:** Video --> Audio
- **Whisper:** Audio --> Text
Timestamps

Step 2: Chunking

- Divide text into small chunks.
- Merge 5 chunks together.

Result: Better context.

Step 3: Embedding

- Combine Title + Text.
- Convert into vectors(embeddings).

Result: Better Understanding.

Step 4: Retrieval

- Convert user query into embedding.
- Use cosine similarity.

Result: Best matching chunk.

Step 5: Answer Generation

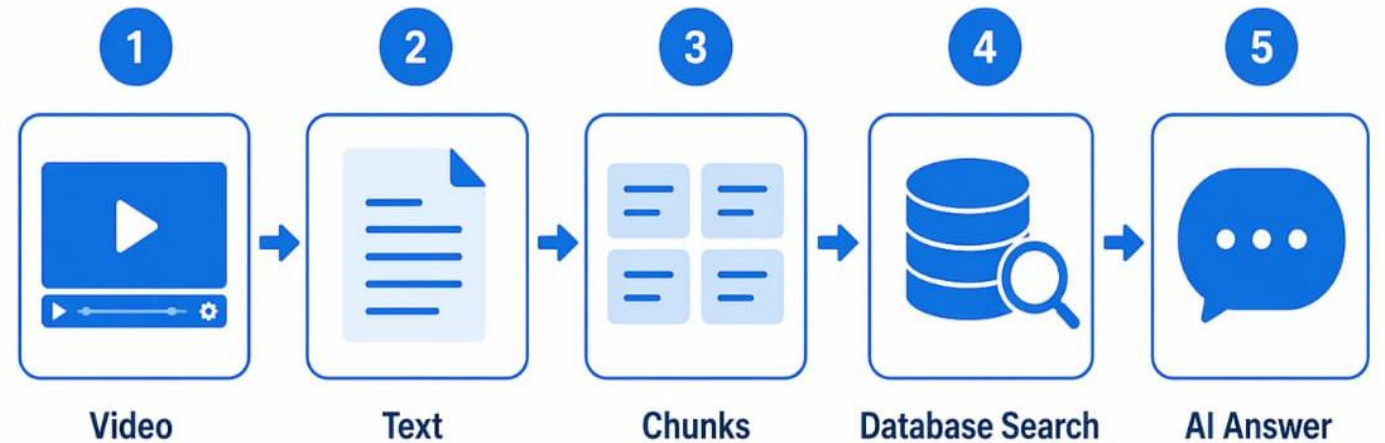
- Use LLM(Llama / GPT)

- Controlled prompt:

No guessing.

Use only given data.

Result: Accurate answer.



Key Improvements:

- Better Context (chunk merging).
- Better retrieval (metadata).
- No hallucinations (prompt control).
- Timestamp-based answers.

Output Example:

Question: What is DataFrame?

Answer: Found at 02:15 – 03:10

Student can directly go to the video.

Advantages:

- Accurate answers.
- Easy to understand.
- Saves time.
- Better Learning experience.

Limitations:

- No full knowledge graph
- Limited deep relationships.

Future Scope:

- Add knowledge graph.
- Improved UI.
- Support multi - answer output.

Conclusion:

- Improved RAG system.
- Better than traditional RAG.
- Useful for Students.

Thank
you...